

Realised Geopolitical Acts and Commodity Stress-Regime Persistence

A time-varying transition-probability hidden Markov approach

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Abstract

We estimate a two-state Gaussian-mixture hidden Markov model for daily Brent crude, gold, silver, and natural gas futures in which transition probabilities depend on lagged geopolitical risk (GPR). Using the threat and act components of the daily GPR index, we find that realised geopolitical acts are associated with greater stress-regime persistence: a one-standard-deviation increase in act elevation is associated with roughly 45 to 51 percent higher odds of remaining stressed. Threat effects are concentrated on entry from calm but are weaker and not robust to the full control set. Nested likelihood-ratio tests reject dropping the geopolitical block from the transition kernel. The results indicate that the geopolitical association is strongest through regime duration rather than through a simple level effect on volatility.

Keywords: Geopolitical risk; Commodity futures; Regime switching; Time-varying transition probabilities; Volatility persistence

JEL classification: C32; C53; G13; Q02; Q43

1. Introduction

Geopolitical events are widely understood to move commodity markets, but their effect is difficult to measure cleanly. Wars, attacks and related threats often arrive alongside exchange-rate moves, inflation news, policy responses and shifts in global risk appetite. The daily Geopolitical Risk (GPR) index of Caldara and Iacoviello (2022) makes this question testable, and recent work has

used it to study commodity connectedness, hedging, resilience and safe-haven volatility (Gong and Xu, 2022; Gupta et al., 2024; Parnes and Parnes, 2025; Yang et al., 2025).

Most of this literature treats geopolitical risk as a level effect: higher GPR is associated with higher commodity prices, returns or volatility. This paper asks a narrower question. Does geopolitical risk change the probability that commodity markets enter, remain in, or exit a stress regime? The distinction matters for risk management. A shock that briefly raises the chance of entering stress has different implications from one that prolongs an episode after stress has already arrived.

A time-varying transition-probability (TVTP) model is suited to that question. In Hamilton's (1989) Markov-switching framework, regime changes are governed by a transition matrix. The TVTP extension of Diebold, Lee and Weinbach (1994) and Filardo (1994) allows those transition probabilities to depend on observed covariates. Placing GPR in the transition kernel therefore lets the model separate entry into stress from persistence within stress. To our knowledge, the threat-act decomposition of GPR has not previously been used in this way to model a common commodity stress regime.

The decomposition is economically important. Threats capture warnings and adverse-event risk; acts capture realised events such as war, attacks and military escalation. These components need not operate through the same margin. Threats may raise uncertainty enough to move a calm market into stress, while realised acts may sustain stress once disruptions have materialised.

The paper contributes by estimating a multivariate commodity HMM in which GPR enters the transition kernel directly rather than as a contemporaneous return or volatility covariate. The main result is an asymmetry between the two GPR components: realised acts raise stress-regime persistence, while threat effects are weaker and concentrated on entry from calm. The design also

deliberately excludes WTI crude oil, whose April 2020 negative settlement is incompatible with the log-return Gaussian-emission structure used below.

The evidence supports a persistence interpretation of geopolitical stress. Realised acts do not simply coincide with volatile commodity returns; elevated act risk is associated with a higher probability that the high-volatility regime continues.

2. Data

We study daily front-month futures settlement prices for Brent crude oil, gold, silver, and natural gas from Bloomberg. The raw sample begins on 3 April 1990, the first date on which all four series are jointly available, and ends on 31 December 2025, spanning 9,327 trading days. WTI crude oil is omitted because its 20 April 2020 negative settlement is incompatible with log returns under the Gaussian-emission structure. The 252-trading-day momentum and roll-yield lookbacks leave an estimation sample of 9,075 observations, beginning in April 1991. The daily GPR index extends back to 1985, so the geopolitical elevation transforms impose no further loss of commodity observations.

Returns are daily log differences of front-month settlement prices. For each commodity we construct roll yield, defined as the log ratio of the front- to second-month contract, and twelve-month-minus-one-month momentum, defined as the log price change from 252 to 21 trading days earlier. We also include the daily log return on the U.S. dollar index. To reduce collinearity, the factor-control specification compresses the commodity controls into four standardised averages: energy and precious-metal curve factors and energy and precious-metal momentum factors.

Geopolitical risk is measured by the daily GPR index of Caldara and Iacoviello (2022) and its threat and act sub-indices. Each series enters as a lagged rolling elevation score: the log level, or $\log l_p$ level for threat and act, less its trailing 252-day mean and divided by its trailing standard

deviation. A value of one therefore denotes GPR one standard deviation above its recent norm. The geopolitical inputs are lagged one trading day before entering the transition kernel. ADF tests reject a unit root and KPSS tests fail to reject stationarity for each transformed GPR series.

3. Methodology

3.1 Emission and regime structure

Let r_t denote the $d \times 1$ vector of daily log returns, with $d = 4$. Conditional on a latent state $s_t \in \{0, 1\}$, returns are drawn from an M -component multivariate Gaussian mixture. Writing the component density as

$$\varphi(r_t; \mu_{km}, \Sigma_{km}) = (2\pi)^{-d/2} |\Sigma_{km}|^{-1/2} \exp\left\{-\frac{1}{2} (r_t - \mu_{km})' \Sigma_{km}^{-1} (r_t - \mu_{km})\right\} \quad (1)$$

the state-conditional emission density is the mixture

$$f(r_t | s_t = k) = \sum_{m=1}^M \omega_{km} \varphi(r_t; \mu_{km}, \Sigma_{km}), \quad \sum_m \omega_{km} = 1, \quad (2)$$

where ω , μ , and Σ are state- and component-specific mixture weights, means, and covariance matrices. We set $M = 2$. States are ordered by volatility: state 0 is calm and state 1 is stress. The mixture emission allows heavy tails and conditional skewness while retaining a tractable likelihood. Table 1 reports implied annualised state volatility, defined as the square root of the average diagonal state covariance scaled by $\sqrt{252}$.

3.2 Time-varying transitions

Transition probabilities vary with a covariate vector Z_t through a multinomial logit in which the calm state is the reference destination ($\gamma_{j0} = 0$ for all origins j):

$$P(s_t = k | s_{t-1} = j, Z_t) = \exp(\gamma_{jk}' Z_t) / \sum_l \exp(\gamma_{jl}' Z_t), \quad (3)$$

which for $K = 2$ reduces to the binary logistic form $P(s_t = 1 | s_{t-1} = j, Z_t) = \exp(\gamma_j' Z_t) / [1 + \exp(\gamma_j' Z_t)]$, so that γ_0 governs the calm-to-stress (entry) transition and γ_1 the stress-to-stress (persistence) transition. The covariate vector stacks an intercept, the lagged geopolitical variable(s) g_{t-1} , and market controls x_t :

$$Z_t = (1, g_{t-1}, x_t')'. \quad (4)$$

Each geopolitical input is a rolling elevation score: the lagged log level (log1p for the threat and act components) demeaned and scaled by its own trailing 252-day moments,

$$g_{t-1} = (\ell_{t-1} - \bar{\ell}_{t-1}) / \varsigma_{t-1}, \quad (5)$$

where $\bar{\ell}$ and ς are the trailing 252-day mean and standard deviation of ℓ . Because the geopolitical block is lagged before entering the transition from $t-1$ to t , GPR is predetermined relative to the regime transition being modelled. The common stress-tilt specification imposes a single shared geopolitical coefficient across origin states,

$$\gamma_{0,GPR} = \gamma_{1,GPR} = \theta, \quad (6)$$

so that elevated GPR shifts the next-period probability toward stress by the same amount from either origin state. This tests the practitioner claim that geopolitical risk raises stress while using one degree of freedom. The components specification leaves the threat and act coefficients free across origins, allowing each component to load on entry, persistence, or neither. Each model is estimated with factor controls and with the full set of individual roll-yield and momentum controls.

3.3 Likelihood and estimation

Collecting all parameters in Θ , the sample log-likelihood is evaluated by the Hamilton filter. The one-step-ahead predicted state probabilities are

$$P(s_t = k | r_{1:t-1}) = \sum_j P(s_t = k | s_{t-1} = j, Z_t) P(s_{t-1} = j | r_{1:t-1}), \quad (7)$$

and the filtered probabilities update with the period likelihood contribution $f_t = \sum_k P(s_t = k | r_{1:t-1}) f(r_t | s_t = k)$:

$$P(s_t = k | r_{1:t}) = P(s_t = k | r_{1:t-1}) f(r_t | s_t = k) / f_t. \quad (8)$$

The model is estimated by maximising $\ell(\Theta) = \sum_t \log f_t$ with the expectation-maximisation (EM) algorithm. The E-step computes smoothed state probabilities $u_t(k)$, pairwise transition posteriors $v_t(j, k)$, and component responsibilities $\rho_t(k, m)$. The emission M-step is

$$\omega_{km} = \sum_t \rho_t(k, m) / \sum_t u_t(k), \quad \mu_{km} = \sum_t \rho_t(k, m) r_t / \sum_t \rho_t(k, m), \quad (9)$$

$$\Sigma_{km} = \sum_t \rho_t(k, m) (r_t - \mu_{km})(r_t - \mu_{km})' / \sum_t \rho_t(k, m). \quad (10)$$

The transition M-step maximises the responsibility-weighted multinomial-logit objective with a ridge penalty ($\lambda = 1$) on standardised slope coefficients, leaving intercepts unpenalised. The common-tilt restriction is imposed inside this step and solved by L-BFGS-B:

$$\hat{\gamma} = \arg \max_{\gamma} \sum_{t \geq 2} \sum_{j,k} v_t(j, k) \log P(s_t = k \mid s_{t-1} = j, Z_t) - (\lambda/2) \|\gamma_{\text{slope}}\|^2. \quad (11)$$

Each fit uses ten random initialisations, retaining the converged solution with the highest log-likelihood. The likelihood spread across restarts is small relative to the level of the objective, and states are canonicalised by volatility after convergence.

3.4 Inference

Inference uses transition-block Hessian standard errors, nested likelihood-ratio tests, and a fixed-block bootstrap. The Hessian standard errors condition on the emission parameters and are therefore treated as lower-bound uncertainty estimates. Coefficients are accompanied by odds ratios, $\exp(\gamma)$, which give the multiplicative change in the odds of the stress destination per standard deviation of elevation. Nested LR tests compare each augmented kernel with a restricted kernel that drops the geopolitical block and is re-estimated on the same sample,

$$\text{LR} = 2 [\ell(\hat{\Theta}_{\text{full}}) - \ell(\hat{\Theta}_{\text{restr}})] \rightarrow \chi^2_q, \quad (12)$$

with q equal to the number of geopolitical restrictions. Following Künsch (1989), the fixed-length block bootstrap uses 63-trading-day blocks and 500 successful replications, re-estimating the full model with ten random initialisations in each replication.

4. Results

The fitted regimes are well separated. Annualised return volatility averages about 27.5 percent in the calm state and 58 percent in the stress state across the four specifications (Table 1). Both states

are persistent, with the stress regime continuing from one day to the next with probability above three-quarters at average GPR.

Table 1. Estimated regime volatilities (annualised, %).

Specification	Calm (S0)	Stress (S1)	Obs.
Common tilt, factor	27.0	57.3	9,075
Common tilt, full	27.5	58.0	9,075
Components, factor	27.4	57.5	9,075
Components, full	27.7	58.1	9,075

Notes: Annualised return volatility implied by the fitted state covariance matrices. The estimation sample comprises 9,075 trading days, running from April 1991 (after the 252-trading-day control burn-in) to 31 December 2025; the underlying data begin on 3 April 1990.

The common stress-tilt coefficient is positive in both control sets. It equals 0.136 (bootstrap SE = 0.069, $p = 0.047$) under factor controls and 0.153 (bootstrap SE = 0.075, $p = 0.041$) under full controls, implying a 15 to 17 percent increase in the odds of stress per standard-deviation of GPR elevation (Table 2). The nested LR test rejects the restricted transition kernel (factor controls: LR = 9.88, $df = 1$, $p = 0.002$; full controls: LR = 7.51, $df = 1$, $p = 0.006$).

The components specification shows that the common tilt is mainly an act-persistence effect. Under full controls, the act-elevation coefficient on the stress-to-stress transition is 0.409 (bootstrap SE = 0.160, $p = 0.011$), an odds ratio of 1.51. The factor-control estimate is similar at 0.371 (bootstrap SE = 0.132, $p = 0.005$, odds ratio 1.45). By contrast, the act coefficient on entry from calm is close to zero in both specifications. Threat-entry evidence is weaker: the threat coefficient from calm is marginal under factor controls (0.191, $p = 0.087$) and not significant under full controls (0.125, $p = 0.252$), while the threat-persistence coefficient is insignificant in both models. The joint LR test rejects dropping the threat-act block (factor: LR = 12.80, $df = 4$, $p = 0.012$; full: LR = 11.02, $df = 4$, $p = 0.026$).

Table 2. Geopolitical transition coefficients.

Coefficient	Transition	Estimate	Bootstrap SE	Bootstrap p	Odds ratio
Panel A: Common stress-tilt					
GPR (factor)	common	0.136	0.069	0.047**	1.15
GPR (full)	common	0.153	0.075	0.041**	1.17
Panel B: Threat / act components					
Threat elevation (factor)	S0→S1 (entry)	0.191	0.111	0.087*	1.21
Act elevation (factor)	S0→S1 (entry)	-0.026	0.115	0.821	0.97
Threat elevation (factor)	S1→S1 (persist.)	-0.163	0.202	0.419	0.85
Act elevation (factor)	S1→S1 (persist.)	0.371	0.132	0.005***	1.45
Threat elevation (full)	S0→S1 (entry)	0.125	0.109	0.252	1.13
Act elevation (full)	S0→S1 (entry)	-0.043	0.120	0.723	0.96
Threat elevation (full)	S1→S1 (persist.)	-0.056	0.184	0.761	0.95
Act elevation (full)	S1→S1 (persist.)	0.409	0.160	0.011**	1.51

Notes: Estimates are transition-kernel coefficients on standardised geopolitical elevation scores. Standard errors and p-values are from the fixed-block bootstrap using 63-trading-day blocks, 500 successful replications, and ten random initialisations per bootstrap replication. For the common stress-tilt models, the common coefficient is imposed across origin states. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 3 reports fitted one-step transition probabilities from the common-tilt factor specification. At average GPR, the stress regime persists with probability 0.79; at two standard deviations above average, this rises to 0.83. The probability of entering stress from calm rises more modestly, from 0.074 to 0.094. The implied expected length of a stress state rises from roughly 3.9 trading days at two standard deviations below average GPR to about 5.9 trading days at two standard deviations above it.

Table 3. Fitted one-step transition probabilities by geopolitical elevation (common tilt, factor controls).

GPR elevation	P(stress calm)	P(stress stress)
-2 SD	0.058	0.743
-1 SD	0.065	0.767
Mean	0.074	0.790
+1 SD	0.084	0.811
+2 SD	0.094	0.831

Notes: Probabilities are evaluated at the fitted coefficients for the common-tilt factor specification with all non-geopolitical covariates held at their sample means.

Two qualifications are important. The Hessian standard errors condition on fixed emissions, so Table 2 reports fixed-block bootstrap standard errors and p-values as the main coefficient evidence. The act-persistence coefficient survives bootstrap resampling in both control sets. The

threat-from-calm coefficient is only marginal under factor controls and is not robust under full controls.

5. Discussion and conclusion

The asymmetry we document admits a natural economic interpretation. Threats are anticipatory information: a credible warning of conflict raises uncertainty and may dislodge a calm market into a high-volatility state, but a threat need not involve a realised material disruption and may therefore exert less force once stress has already arrived. Realised acts are different in kind. An attack, blockade, or other realised act corresponds to a materialised and potentially ongoing disruption to supply, transit, digital access, or expectations, and is therefore more consistent with sustaining the stressed configuration than creating it. The two components of geopolitical risk therefore point to different margins of the transition kernel, but the robust margin is act-driven persistence.

The finding carries a practical implication for risk management. If realised acts prolong stress, then exiting a geopolitically driven volatility episode is governed less by the level of rhetoric than by whether materialised events continue to sustain disruption. A fall in threat intensity need not shorten an episode that realised events are actively sustaining, and hedges calibrated to the level of geopolitical risk may understate the duration of the regimes that realised acts produce.

Several limitations bound the claims. The analysis is in-sample and descriptive of the joint dynamics of returns and geopolitical risk; it is not a tradeable out-of-sample forecast, which we leave to companion work. The threat and act elevation series are positively correlated by construction, so the individual decomposition should be read alongside the joint likelihood-ratio evidence rather than in isolation, and the negative threat-on-persistence point estimate may partly reflect this collinearity. Subject to these qualifications, the evidence indicates that realised geopolitical acts are associated with commodity stress primarily through persistence, and that the

geopolitical effect identified here is stronger for the persistence of volatility regimes than for their onset.

Declarations

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Data availability: The Caldara–Iacoviello geopolitical risk index is publicly available. Futures price data are licensed from Bloomberg and cannot be redistributed; replication code and the constructed transition-variable panel will be made available in a public repository on acceptance.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process: During preparation of this work the author used OpenAI ChatGPT to support language editing and manuscript structure. The author reviewed and edited the content and takes full responsibility for the manuscript.

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